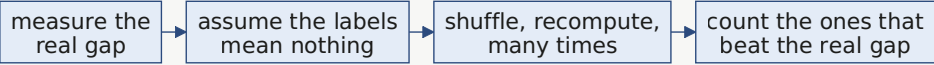


Real or noise, cause or coincidence

Week 1 Day 4, Kalpa Retail. Three questions that look alike and need three different habits. The page to keep for every interview that asks whether a number means anything.

1 The shuffle, in one picture



That count, over how many worlds you made, is the p-value. Ten playing cards do it; the code does it five thousand times.

Crux. The unit you shuffle is the customer and not the order, because a customer's orders belong together and splitting them builds a world that could not exist.

2 What a p-value is not

It is	It is not
The share of chance-only worlds at least this extreme	The chance the finding is wrong
A statement about the null world	The chance that chance caused it
Silent about size	A measure of how big the effect is
The first wrong reading is the one said aloud in meetings, and it reverses the meaning.	

3 Never write p equals zero

Five thousand shuffles resolve to one in five thousand. Write $p < 0.0002$.

The number you report is the resolution of your own simulation, not a certainty your method cannot produce.

4 Two results, opposite verdicts

	Retail-Plus	Student
Seen	a 32.3 point gap	5 orders then 7
Extreme shuffles	0 of 5,000	1,914 of 5,000
Reported as	$p < 0.0002$	$p = 0.383$
Reads as	chance does not do this	chance does this two in five
Crux. Distrust any rate on fewer than about thirty observations, and say it is a rule of thumb rather than a law when you use it.		

5 Three separate calls

Question	Answered by
Could chance have done this?	The p-value
Is it big?	The size of the effect
Is it worth acting on?	The cost against the gain
Only the first comes from the data. A tiny p-value on a tiny effect is a precise measurement of something that does not matter.	

6 A fair comparison

A group that did **not** get the thing, that is **like** the group that did, in the ways that matter.

Targeting breaks the second half on purpose. The monsoon sale went to Retail-Plus, who already spend two and a half times more, so the exposed group was richer before the discount did anything.

Crux. When a comparison reverses once you split the groups, the aggregate was being driven by the mix rather than by the thing you changed.

7 The note, four parts

Part	Carries
Claim	One sentence, with the number and its denominator
Evidence	What you computed, and on how many
Caveat	The thing that would change the claim
Action	What to do, and what it costs
The sentence you are paid for: "We do not know yet, and here is what would tell us."	